CHAPTER 12

Parametric representations and equations Cartesian

S65, S67, S70

Before you start · Rights, plans and systems

Without a calculator, in six minutes. Respond briefly. Mobilize guiding vectors, vectors and membership. Then choose the targeted review indicated under the questions.

1. Provide a guiding vector of (AB) for A=(1;2;3), B=(3;1;3).

2. Verify that (1-2;-1) is orthogonal at (2;-1;0) and (1;0;1).

3. Is the vector (1;1;1) orthogonal at (1;2;-1)?

4. Resolve 2 $t = 6$.

5. What needs to be checked for a point to be projected M on a plan?

6. Are vectors (1.0;0) and (0;1;0) colinear?

Responses

$(2; -1; 0)$; both products are worth 0 ; no, the product is worth 2 ; $t = 3$; belonging to the plan and orthogonality; no, no multiple of the first gives the second.

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Subtract A to B coordinated by co-ordinate; R05. **2.** Calculate the two scalar products separately; R60. **3.** Add the three products and compare to zero; R60. **4.** Divide the two members by 2 and Check. **5.** Resuming the two distinct conditions of the project; R63. **6.** Compare the first then the second coordinates; R06. R-questions serve as support for review with the Professor, even if their collective ritual is planned later.

Objectives

Describe a line and a plane; determine intersections; choose a method for a projection.

Course notes

In all this chapter, the coordinates are given in an orthonormal coordinate system of space. hypothesis is necessary to identify a normal vector from the coefficients of an equation and to calculate distances.

Adjusted lines

The right through $A(x_A, y_A, z_A)$ of a non-zero master vector (a, b, c) admits representation

$$d : \begin{cases} x = x_A + at \\ y = y_A + bt \\ z = z_A + ct \end{cases} \quad (t \in \mathbb{R}).$$

A point belongs to the right if it exists the same real t a right admits of many performances: change of starting point or multiplication of the director by a real non-zero.

Example. $(x, y, z) = (1, 2, 0) + t(2, -1, 3)$ describes the same line as $(3, 1, 3) + s(4, -2, 6)$; the link is $t = 1 + 2s$.

Cartesian plans

Equation of a plan

If $(a, b, c) \neq (0, 0, 0)$, the whole $ax + by + cz + d = 0$ is a normal vector plane (a, b, c). The normal plan $\vec{n}$ passing through A is built by writing $\overrightarrow{AM} \cdot \vec{n} = 0$.

Demonstration. The condition $\overrightarrow{AM} \cdot \vec{n} = 0$ written $a(x - x_A) + b(y - y_A) + c(z - z_A) = 0$. By developing, $d = -ax_A - by_A - cz_A$. Conversely, any satisfactory point this equation verifies the condition of orthogonality, therefore belongs to the plan.

Example. The normal plan $(2, -1, 3)$ passing through $A(1, 2, 0)$ is 2 $(x - 1) - (y - 2) + 3z = 0$, either $2x - y + 3z = 0$.

Intersections

For a right and a plan, replace x, y, z in the plane equation by their expressions parametric. A solution in t gives a point intersection; an identity gives a right included; a contradiction gives a disjointed parallel line. For two planes, normals do not colinears give an intersection on a straight line; solve the system with two equations in Choosing a variable as a parameter. If normals are collinear, the planes are parallel: Distinguished and disjointed. For two straight lines, use parameters s, t independent and solve the three equations. Without solution and with non-collinear directors, the straight ones are not coplanar.

Projection on a plan

If $\mathcal{P} : ax + by + cz + d = 0$, look for the planned M on the normal right passing through M : $H = M + t(a, b, c)$. Replace in the plan equation to determine t, then calculate MH. - We're in. deduced, without making it a formula to be learnt in isolation,

$$d(M, \mathcal{P}) = \frac{|ax_M + by_M + cz_M + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Verification is essential. After calculating a projected, check simultaneously $H \in \mathcal{P}$ and the colinearity of $\overrightarrow{MH}$ with normal.

Moving from objects to equations

12.1 One straight line, several settings

First use: S65.

Definition · Set a straight line

A line passing through A and a non-zero steering vector $\vec{u}$ is all M points such as that $\overrightarrow{AM} = t\vec{u}$, $t \in \mathbb{R}$. The parameter is neither a co-ordinated nor a time, unless the context specifies it.

Example resolved · Same set, other parameters

Expressions $(1 + t; 2 - t; 2t)$ and $(3 + 2s; -2s; 4 + 4s)$ describe the same line: pose $t = 2 + 2s$ transforms the first into the second, and this transformation travels through all the real. The starting points may be different, as may proportional directors.

A1 · Immediate application Show that $(t; 1 + 2t; 3 - t)$ and $(2 + 3s; 5 + 6s; 1 - 3s)$ describe the the same right. $(1; 3; 2)$ Does he belong there?

12.2 A normal fixed plane equation

First use: S67.

Property · Cartesian Equation

In an orthonormal coordinate system, a normal non-zero plane $(a; b; c)$ passing through $A = (x_A; y_A; z_A)$ a for equation $a(x - x_A) + b(y - y_A) + c(z - z_A) = 0$. It translates $\overrightarrow{AM} \cdot \vec{n} = 0$.

Example solved · Build and then test

For A=(1;0;2) and $\vec{n} = (2; -1; 3)$, the plan has for equation 2 $x - y + 3z - 8 = 0$. Item (0;1;3) belongs to the plan because $0 - 1 + 9 - 8 = 0$. A point is not in the plan because its triplet looks like the normal vector.

A2 · Partial guidance Determine the normal plan $(1; 2; -1)$ Test membership through (2.0;1). of (1;1;2), then give two non-collinear directions of the plan.

12.3 Three exits for one equation

First use: S69.

After substitution	Interpretation	Example
$at + b = 0$, $a \neq 0$	one point	straight stinging
$0t + b = 0$, $b \neq 0$	no point	strict parallel
$0t + 0 = 0$	all points	right included

Example Solved · Read a reduced equation

For the plane x+y+z=1 and the right $(t;2t;1-t)$, substitution gives 2 $t+1=1$, so t=0 and a single intersection (0;0;1). It is concluded only after reinjecting the value in the three coordinates.

A3 · Compare In the plan $z=1$, to study the positions of the right $d_1:(t;0;t)$, $d_2:(t;0;0)$ and $d_3:(t;0;1)$. Give the intersection point when it is unique and outline the three situations.

12.4 Two straight lines, two independent parameters

First use: S70.

Solved Example · Do not impose false simultaneity

For $d:(t;0;0)$ and $d':(1;1+s;0)$, the intersection requires t=1 and s=1. Imposing t=s would disappear Wrongly the point (1.0;0). The parameters are used separately to navigate each right.

A4 · Repair a Method For $d:(t;t;0)$ and $d':(1+s;1-s;s)$, solve with two parameters, then explain why the three coordinates should be checked.

12.5. Two verifications after the system

First use: S70.

Example Solved · Project on a Plan

Let $\mathcal{P}:x+2y-z=1$ and M=(1;1;1). Normal per M is $H=(1+t;1+2t;1-t)$. Replace in the plan gives 2 $+6t=1$, therefore $t=-1/6$. H=(5/6;2/3;7/6). $/6+4/3-7/6=1$ and $\overrightarrow{MH}=(-1/6)(1;2;-1)$. Distance is $\overline{6}/6$.

A5 · Synthesis with degenerate cases For a real m, d_m passes through (0;0;m) and has for direction $(1;m;1)$. Studying the intersection with $x+2y-z=1$. Distinguish m=0. Is there m for which the right is perpendicular to the plane?

A first step: Replace the three coordinates before dividing. For orthogonality, compare the director to A normal one.

A6 · Choice of strategy Determine the intersection of the planes $x+y+z=3$ and $x-y=0$. Provide one point and a guiding vector, then control these two informations in the equations.

Wait a minute at the courthouse. State the two properties that allow to recognize a project without indicate coordinates.

Review postponed · S71 and then S87 Match each line of an algebraic calculation to a geometric property; treat first the parameters that cancel a denominator.

Research activity

Three straight lines in front of a plan · 30 min.

In an orthonormal space coordinate system, the plane is considered $\mathcal{P}$ (d) equation $x+y+z=3$. In each question, the parameter t travel $\mathbb{R}$.

1. The right d_1 admits representation $(x;y;z)=(t;t;t)$. Determine $d_1\cap\mathcal{P}$.

2. The Right d_2 admits representation $(x;y;z) = (3+t;-t;0)$. Determine $d_2 \cap \mathcal{P}$.

3. The right d_3 admits representation $(x;y;z) = (t;-t;0)$. Determine $d_3 \cap \mathcal{P}$.

4. For each line, specify whether the equation obtained after substitution in the plane has a solution unique, no solution or any real solution. Linking these cases to geometric positions found.

Progressive exercises

E1 · Direct application. In a space coordinate system, we consider $A(1;-1;2)$ and $B(3;0;1)$. Provide a parametric representation of the right (AB).

E2 · Application and justification. In an orthonormal coordinate system, determine a Cartesian equation of the plan through $A(1;2;3)$ and normal vector $\vec{n} = (1;0;-2)$.

E3 · Method to be organised. In a space coordinate system, the right d is given by $(x;y;z) = (t;1+t;2-t)$, $t \in \mathbb{R}$, and plan $\mathcal{P}$ by $x+2y+z=8$. Determine their intersection.

E4 · Choice of strategy. In a space coordinate system, determine the intersection of the planes of equations $x+y+z=1$ and $x-y=0$. Provide a parametric representation if the intersection is a straight line.

E5 · Synthesis or demonstration. In a space coordinate system, study the relative position of the planes d-equations 2 $x-y+z=3$ and $4x-2y+2z=5$. Justify.

Work in the manual or annal

Baccalaureate, Métropole, 17 June 2025: Exercise 3, assertions 1 to 4 (p. 5·6). At home after S70. Justify each answer; for a false statement, explain precisely what refutes it.
subject is available here: official annual . Use the data and instructions of this yearnal.

Duty at home

DM12 · Projection and intersection · 55 min.

In an orthonormal space coordinate system, the point is considered $M(3;0;0)$ and plans $\mathcal{P} : x+2y+2z=9$ and $\mathcal{Q} : x-y=0$.

1. Give the coordinates of a normal non-zero vector to $\mathcal{P}$.

2. Determine coordinates of the orthogonal project H of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$.

3. Show that $\mathcal{P} \cap \mathcal{Q}$ is a straight and give a parametric representation.

4. Item H Does it belong to this right?

5. Determine coordinates of the point K of this right whose rating is zero. MKthen compare MK and MH.

QCM A · read and check a representation

Orthonormal coordinate system. After S67; QCM B after S70.

For each question, only one proposal is accurate.
consult the correction.

1. $d : (x; y; z) = (1; 0; 2) + t(2; -1; 0)$ admits as director:

A. $(1; 0; 2)$ B. $(2; -1; 0)$

C. $(2; 1; 0)$ D. $(0; 0; 2)$

2. Which of these points belongs to d ?

A. $(3; 1; 2)$ B. $(1; 0; 0)$

C. $(2; -1; 0)$ D. $(3; -1; 2)$

3. A normal vector on the plane $x + 2y - 2z = 3$ is:

A. $(1; 2; -2)$ B. $(1; 2; 3)$

C. $(3; 0; 0)$ D. $(-2; 1; 0)$

4. To check H Projected M on $\mathcal{P}$, it must:

A. Only $H \in \mathcal{P}$ B. Only MH small

C. $H \in \mathcal{P}$ and $\overrightarrow{MH}$ normal on the plan D. That $H = M$

5. Right $A + tu$ with $u \neq 0$ is also represented by:

A. $A + t\vec{0}$ B. $A + 2u + s(3u)$

C. $tA + u$ D. the only point $A + u$

MCQ B · intersections and special cases

For each question, only one proposal is accurate.
consult the correction.

6. The substitution of a line in a plane equation gives 0 $= 4$. The intersection is:

A. a right B. One point

C. the plan D. empty

7. If substitution gives 0 $= 0$ for any parameter:

A. the right is included in the plane B. there is no common ground

C. There is only one point in common D. The data are incorrect

8. To search for the intersection of $A + tu$ and $B + sv$We have to:

A. Impose $s = t$ B. Replace both parameters with 0

C. Solve for s, t Independent D. ignore the third co-ordinate

9. The intersection of $x + y = 1$ and $z = 2$ is:

A. a plan B. Right $(t; 1 - t; 2)$

C. the point $(1; 0; 2)$ D. empty

10. For $m = 0$, total $mx + y = 1$ is:

A. Empty B. a right

C. All space D. The plan $y = 1$

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. Two straight lines which have non-colonial directors are necessarily stinging.

2. Plans $x + y + z = 1$ and $2x + 2y + 2z =$ Three are confused.

3. The proposed M on a plan is unique.

Corrected MCQs

1. **B.** The multiplication vector t is a director; the first triplet designates a point on the right.

2. **D.** Take $t = 1$ simultaneously fulfils the three coordinates. Only one correct coordinate does It's not enough.

3. **A.** In an orthonormal coordinate system, the coefficients of x, y, z give a normal vector. constant is not its third coordinated.

4. **C.** These two conditions characterize the project. M is in the plane, the null vector is suitable and $H = M$.

5. **B.** The new point is on the right, and 3 u is a non-zero director colinear to u. Change $t = 2 + 3s$ travels all $\mathbb{R}$.

6. **D.** No value of the parameter is suitable: the line is strictly parallel to the plane.

7. **A.** Each point on the right checks the equation of the plane. The intersection assembly is all the Right.

8. **C.** The same point can be reached for two different values of the parameters. must be checked.

9. **B.** Both constraints leave a free parameter; take $x = t$ imposes $y = 1 - t$ and $z = 2$.

10. **D.** The coefficient of y An equation can still define a plane when a coefficient is zero.

True/false. False: they can be non-coplanar. False: they have normal colinear but Incompatible constants; the planes are strictly parallel. True: the normal straight from M Cut the plan into one point.

Corrected applications and support

Application 1. Set t=2+3s; substitution gives the three coordinates of the second setting and travels $\mathbb{R}$. For (1;3;2), t=1 satisfies the three coordinates.

Application 2. Equation $x+2y-z=1$. The proposed item gives 1 $+2-2=1$, therefore belongs to the possible directions $(2;-1;0)$ and $(1;0;1)$, orthogonal to normal and non-collinear.

Implementation 3. d_1 cut in (1.0;1); d_2 is strict parallel; d_3 is included. t=1, 0=1 and 1=1, respectively.

Implementation 4. The equations are t=1+s, t=1-s and 0=s. So s=0, t=1, point (1;1;0). Solve two equations can produce candidates; only the third check concludes.

Application 5. $d_m : (t; mt; m+t)$. The equation becomes $2mt-m=1$. If m=0, it is impossible: straight strict parallel. Otherwise $t=(m+1)/(2m)$ provides a single point. To be perpendicular, $(1;m;1)$ (1.2;-1); the first co-ordinated requires a ratio 1, incompatible with the Third. No m.

Implementation 6. x=y=t and z=3-2t, so straight $(t;t;3-2t)$, point (0;0;3), direction (1;1;-2). meets both equations; the director is orthogonal to both normal (1;1;1) and (1;-1;0).

From system to geometry

To predict the three cases using the figure, then read equations 0=1 and 0=0 as useful results. To differentiate, provide the substitution to the student who mastered the idea but gets lost in the calculation.

Conduct and correction

Benchmarks for the teacher

Use two different parameters to compare two lines. Check the three coordinates, even if two are sufficient to calculate candidates. A Cartesian equation alone generally describes a plan in space; it does not describe a right. Reinvest the projected chapter 11 for give meaning to system resolution.

Corrected activity

1. A point of d_1 a the coordinates (t,t,t). It belongs to the plan if and only if 3 $t=3$, either $t=1$. The intersection is therefore the only point $(1,1,1)$.

2. For a point of d_2, $x+y+z=3+t-t+0=3$ whatever t. All points on the right are in the plan: $d_2 \subset \mathcal{P}$.

3. For a point of d_3, $x+y+z=t-t+0=0$. Equation $0=3$ has no solution: the right is dissociated from the plan. $(1,-1,0)$ is orthogonal to normal $(1,1,1)$; it is parallel to Plan.

4. After substitution, one obtains an equation with one solution, one identity, respectively. and a contradiction. These three cases correspond to an intersection a point, inclusion and absence of intersection.

E1. $\overrightarrow{AB}=(2,1,-1)\neq\vec{0}$. The right has therefore as representation $(x,y,z)=(1,-1,2)+t(2,1,-1)$, $t\in\mathbb{R}$. Values $t=0$ and $t=1$ give again A and B.

E2. For $A(1,2,3)$, condition $\overrightarrow{AM}\cdot(1,0,-2)=0$ is $(x-1)-2(z-3)=0$. After development, the plan has for equation $x-2z+5=0$. Item A Check it out.

E3. After substitution, $t+2(1+t)+(2-t)=8$ or 2 $t+4=8$, so $t=2$. The only intersection is $(2,3,0)$; check 2 $+2\times3+0=8$.

E4. The second equation gives $x=y$. Set $x=y=t$; the first becomes 2 $t+z=1$. All solutions is $(x,y,z)=(t,t,1-2t)=(0,0,1)+t(1,1,-2)$, a right.

E5. The normals are colinear, the second being double the first. $x-y+z=3$ gives 4 $x-2y+2z=6$, incompatible with the second member 5. The plans are therefore strictly parallel, and non-separating.

Corrected from DM12

1. The coefficients x, y, z give the normal non-zero $\vec{n} = (1, 2, 2)$ in the orthonormal coordinate system.

2. Let's look. H on the normal passing through M : $H = (3 + t, 2t, 2t)$. Membership of $\mathcal{P}$ gives $3 + t + 4t + 4t = 9$, so $t = \frac{2}{3}$. Thus $H = (\frac{11}{3}, \frac{4}{3}, \frac{4}{3})$. We're checking. $\frac{11}{3} + \frac{8}{3} + \frac{8}{3} = 9$ and $\overrightarrow{MH} = (\frac{2}{3})\vec{n}$. So H is the projected, and $MH = (\frac{2}{3})\sqrt{1 + 4 + 4} = 2$.

3. The Normal $(1, 2, 2)$ and $(1, -1, 0)$ are not collinear: the plans cut on a straight line. Like $x = y$, posisons $x = y = s$. The first equation gives 3 $s + 2z = 9$, either $z = \frac{9-3s}{2}$. The intersection is the right $(x, y, z) = (s, s, \frac{9-3s}{2})$, $s \in \mathbb{R}$.

4. Item H a $x_H = \frac{11}{3} \neq \frac{4}{3} = y_H$, so it does not belong to $\mathcal{Q}$ or to the right of intersection.

5. Rating is the co-ordinate z. Its nullity requires $9 - 3s = 0$, therefore $s = 3$ and $K = (3, 3, 0)$. Then $MK = 3 > 2 = MH$. This result is consistent with the property of the project: among all points of the plan, H is closest to M.

Corrected work in the manual or an anal

1. True: the right passes through B and its policy vector $(-2; -1; 3)$ value $-\frac{1}{2}\overrightarrow{AB}$.

2. False: $\vec{n} \cdot \overrightarrow{OB} = 15 - 4 - 1 = 10 \neq 0$.

3. False: for $k = -4$ and $s = \frac{5}{2}$, both representations give $(11; 12; -14)$. The rights are cut, so they're coplanar.

4. True: $d(C, \mathcal{P}) = \frac{|2-(-1)+2+1|}{\sqrt{1+1+1}} = 2\sqrt{3}$.